CH 4 - Multivariate Discrete Random Variable

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4.1 Basic Terminology and Techniques

Ofter we are interested in multiple phenomenal that vary alongside each other.

Examples:

- Treatment/dose and disease status
- Stock price or credit rating of 2 companies
- Pollutant level and respiratory illnesses

> 📘 **Definition — Joint Probability Mass Function**
>
> The **joint probability mass function** of two random variable X and Y is
>
> $$f(x,y) = P(X = x, Y = y)$$
>
> for (x, y) in joint rnage of X, Y respectively.
>
> It is represented usually by a table, occasionally a function, never a histogram.

Properties for JPMF:

1. $0 \leq f(x,y) \leq 1$
2. $\sum_x \sum_y f(x,y) = 1$

We can find any probability involving x and y by summing up the $f(x,y)$ over all pairs that satisfy

Example:

Flip a fair coin 3 times and let $X = \#$ of heads obtained, $Y = \begin{cases} 1 \text{ if first flip is T} \\ 0 \text{ else} \end{cases}$. Find $f(x,y)$

	0	1	2	3
0	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$
1	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0

$P(2 < X + Y) = f(3,0) + f(2,1) + f(3,1) = \frac{2}{8}$

$P(X - Y = 1) = f(1,0) + f(2,1) = \frac{2}{8}$

> 📘 **Definition — Marginal PMF**
>
> The marginal PMF of X is
>
> $$f_X(x) = P(X = x) = \sum_y f(x,y)$$
>
> $$f_Y(y) = P(Y = y) = \sum_x f(x,y)$$

In general, the JPMF of $X_1, X_2, ..., X_n$ is $f(x_1, x_2, ..., x_n) = P(X_1 = x_1, X_2 = x_2, ..., X_n = x_n)$

Marginal PMF of X_1 is $f_1(x_1) = \sum_{X_2} ... \sum_{X_n} f(x_1, x_2, ..., x_n)$.

The JPMF of X_1, X_2 is $f_{1,2}(x_1, x_2) = \sum_{X_3} ... \sum_{X_n} f(x_1, x_2, ..., x_n)$

> 📘 **Definition — Independence Random Variable**
>
> We say two random variables X, Y are independent if and only if
>
> $$f(x, y) = P(X = x, Y = y) = P(X = x)P(Y = y) = f_X(x) + f_Y(y)$$

Example:

Let $X = \#$ days between texts from A to B and $Y = \#$ days between texts from B to A

The joint PMF be $f(x, y) = \frac{1}{16}(\frac{1}{2})^x(\frac{7}{8})^y$, for $x, y = 0, 1, 2, ...$

Find the marginal PMF:

$f_X(x) = \frac{1}{16}(\frac{1}{2})^x \sum_y (\frac{7}{8})^y = \frac{1}{16}(\frac{1}{2})^x 8 = (\frac{1}{2})^{x+1}$

$X \sim \text{Geo}(\frac{1}{2})$

$f_Y(y) = \frac{1}{16}(\frac{7}{8})^y \sum_x (\frac{1}{2})^x = \frac{1}{16}(\frac{7}{8})^x 2 = \frac{1}{8}(\frac{7}{8})^y$

$Y \sim \text{Geo}(\frac{1}{8})$

$f(x, y) = f_X(x)f_Y(y) = (\frac{1}{2})^{x+1}\frac{1}{8}(\frac{7}{8})^y = \frac{1}{16}(\frac{1}{2})^x(\frac{7}{8})^y$, thus X, Y are independent

4.2 Functions of 2^+ random variables

We may be interested in $U = g(X, Y)$

Examples:

1. Let $X = \#$ marks gained, $Y = \#$ marks lost

$y \setminus x$	0	1	2	3
0	0.18	0.34	0.03	0.02
1	0.09	0.09	0.06	0.08
2	0.01	0.01	0.06	0.03

   If U is the overall impact, $U = g(X, Y) = X - Y$

   PMF of U:

u	-2	-1	0	1	2	3
	0.01	0.1	0.33	0.43	0.11	0.02

   In general, if $U = g(X, Y)$, $f_U(u) = \sum_x \sum_y f(x, y)$

   In the case of $U = g(X, Y) = X + Y$

   $f_U(u) = \sum_x \sum_y f(x, y) = \sum_x^u f(x, u - x)$

2. Suppose $X \sim \text{Poi}(7), Y \sim \text{Poi}(5)$ are independent

   $U = X + Y$

   $f_U(u) = f_X(x) f_Y(y) = \frac{e^{-7}7^x}{x!} \cdot \frac{e^{-5}5^y}{y!} = \sum_{x=0}^{u} f(u, u-x) = \frac{1}{e^{12}} \sum_{x=0}^{u} = \frac{7^x 5^{u-x}}{x!(u-x)!}$

   $= \frac{e^{-12}}{u!} \sum_{x=0}^{u} = \binom{u}{x} 7^x 5^{u-x} = \frac{e^{-12}12^x}{u!}$

   $U \sim \text{Poi}(12)$

Poisson random variables are closed under convolution:

$X \sim \text{Poi}(x), Y \sim \text{Poi}(y)$ are independent, $X + Y \sim \text{Poi}(x + y)$

Binomial random variables are closed under convolution:

$X \sim \text{Bin}(n, p), Y \sim \text{Bin}(m, p)$ are independent, $X + Y \sim \text{Bin}(n + m, p)$

Geometric random variables are closed under convolution:

$X \sim \text{Geo}(p), Y \sim \text{Geo}(p)$ are independent, $X + Y \sim \text{NB}(p)$